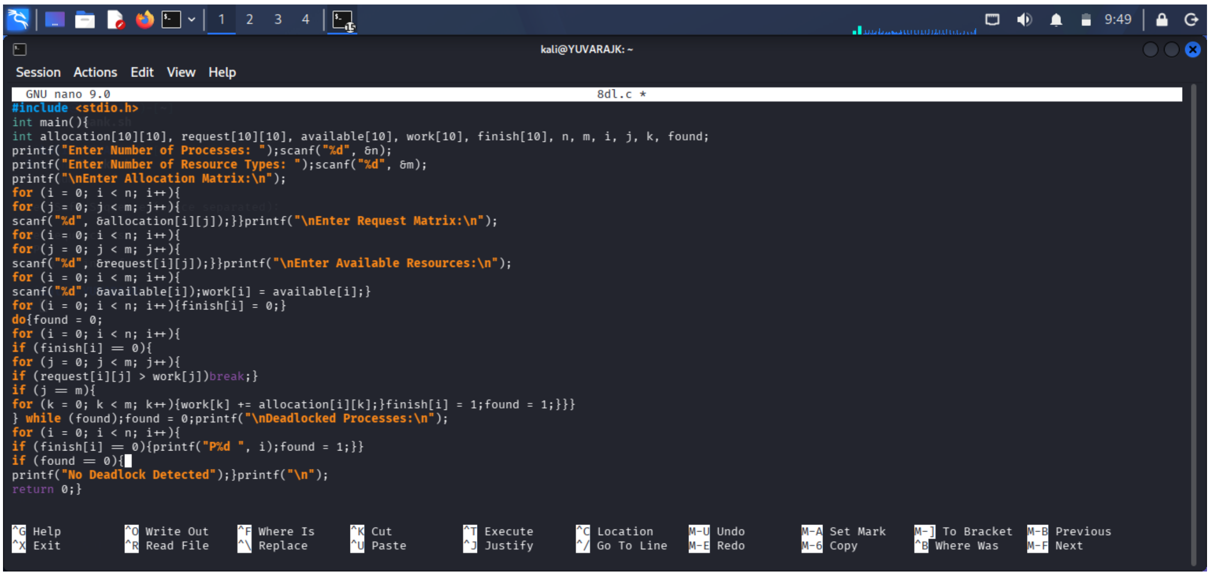
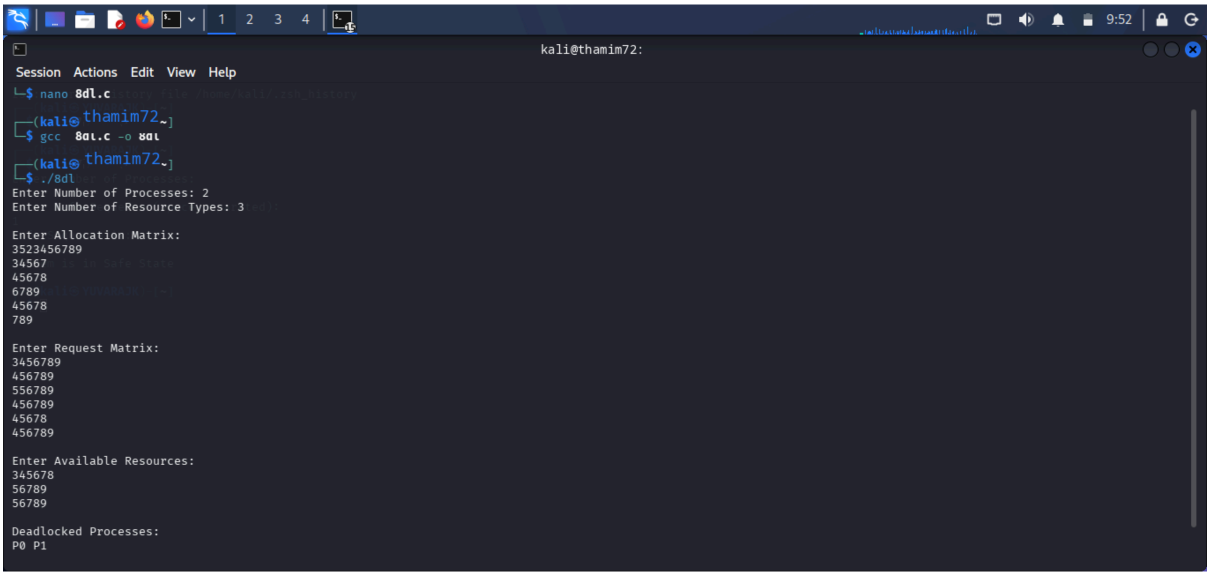
C PROGRAMMING :



```c
#include <stdio.h>
int main(){
int allocation[10][10], request[10][10], available[10], work[10], finish[10], n, m, i, j, k, found;
printf("Enter Number of Processes: ");scanf("%d", &n);
printf("Enter Number of Resource Types: ");scanf("%d", &m);
printf("\nEnter Allocation Matrix:\n");
for (i = 0; i < n; i++){
for (j = 0; j < m; j++){
scanf("%d", &allocation[i][j]);}}printf("\nEnter Request Matrix:\n");
for (i = 0; i < n; i++){
for (j = 0; j < m; j++){
scanf("%d", &request[i][j]);}}printf("\nEnter Available Resources:\n");
for (i = 0; i < m; i++){
scanf("%d", &available[i]);work[i] = available[i];}
for (i = 0; i < n; i++){finish[i] = 0;}
do{found = 0;
for (i = 0; i < n; i++){
if (finish[i] == 0){
for (j = 0; j < m; j++){
if (request[i][j] > work[j])break;}
if (j == m){
for (k = 0; k < m; k++){work[k] += allocation[i][k];}finish[i] = 1;found = 1;}}}
} while (found);found = 0;printf("\nDeadlocked Processes:\n");
for (i = 0; i < n; i++){
if (finish[i] == 0){printf("P%d ", i);found = 1;}}
if (found == 0){
printf("No Deadlock Detected");}printf("\n");
return 0;}
```

OUTPUT :

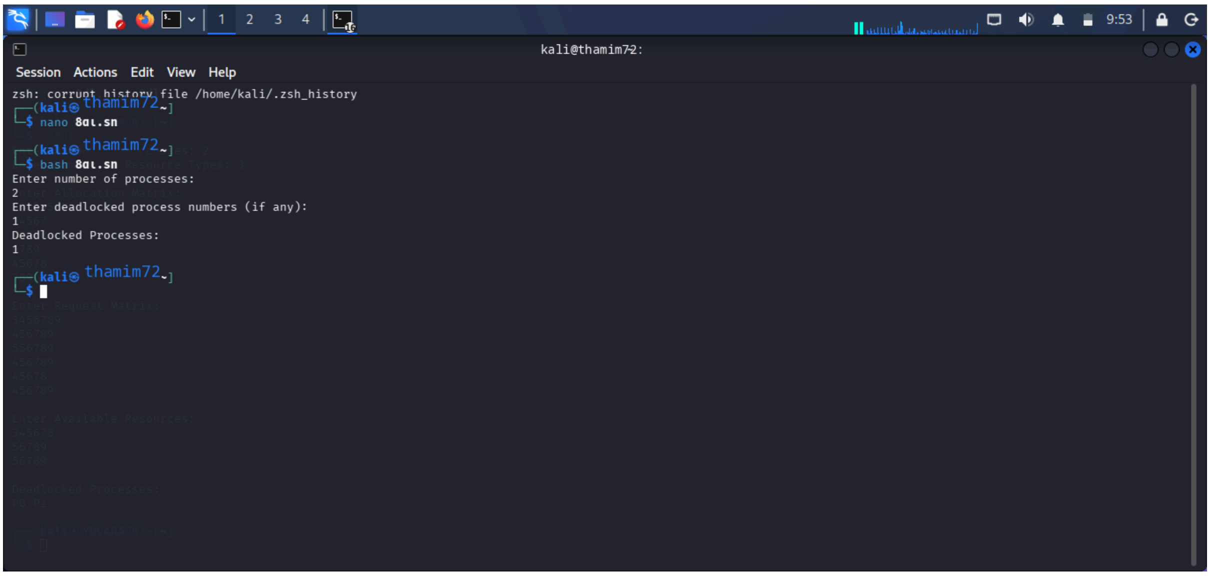


```
└─$ nano 8dl.c

┌──(kali㉿thamim72)-[~]
└─$ gcc 8dl.c -o 8dl

┌──(kali㉿thamim72)-[~]
└─$ ./8dl
Enter Number of Processes: 2
Enter Number of Resource Types: 3

Enter Allocation Matrix:
3523456789
34567
45678
6789
45678
789

Enter Request Matrix:
3456789
456789
556789
456789
45678
456789

Enter Available Resources:
345678
56789
56789

Deadlocked Processes:
P0 P1
```

SHELL PROGRAMMINNG :

```
GNU nano 9.0                         8dl.sh
#!/bin/bash
echo "Enter number of processes:"
read n
echo "Enter deadlocked process numbers (if any):"
read processes
if [ -z "$processes" ]
then
echo "No Deadlock Detected"
else
echo "Deadlocked Processes:"
echo "$processes"
fi
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿thamim72)-[~]
└─$ nano 8dl.sh

┌──(kali㉿thamim72)-[~]
└─$ bash 8dl.sh
Enter number of processes:
2
Enter deadlocked process numbers (if any):
1
Deadlocked Processes:
1

┌──(kali㉿thamim72)-[~]
└─$
```